2018 National Information Technology Conference (NITC), 02-04 October, Colombo, Sri Lanka

Smartphone VPN App Usage and User Awareness among Facebook Users

Asela Jayatilleke
Colombo, Sri Lanka
asela.j89@gmail.com

Parakum Pathirana
Colombo, Sri Lanka
amaranga@gmail.com

***Abstract*—The information security threat landscape is evolving daily with attackers increasingly resorting to 'social engineering' tactics to obtain information about their victims. Such information could be even those published on platforms such as Facebook, which can provide deep insights into users' activities, preferences, likes, dislikes and behavioural patterns. Information of such nature is of great value in today's technology driven society, which is why individuals and organisations are increasingly trying to gather and store them, with or without the consent of the end-user. A common attack vector used for such purposes are mobile apps, primarily due to the fact that smartphones are increasingly capturing data about their users. This research attempts to identify user awareness levels among Sri Lankan smartphone users, when the usage of VPN apps increased suddenly during the period in which social media were banned. Analysis of responses provided by 150 respondents show that 85% admitted to using VPN apps during this period, with only 21.74% of Non-VPN users being aware of the security risks associated with VPN apps.**

Keywords—phishing, social engineering, VPN, user awareness, information security

I. INTRODUCTION

Data is of great value to individuals and more importantly, to organisations. The future of many large-scale organisations, their marketing, operational expansion and investment strategies rely, to a great extent, on information derived from data. Such data may include very simple facts about an everyday, average individual. As such, organisations are investing heavily into new technologies and strategies that can capture and deliver the required data for their analysis.

In today's technology driven society, capturing data from a user's computer can provide deep insights into their complete lifestyles. With the mobile phones becoming 'smarter' and individuals spending more time on their mobile devices, even more data can be captured, all of which provides valuable inputs to business organisations.

Data capturing is usually done with the consent of the person involved, giving them the option to refrain from sharing. However, some data gathering techniques involve secretly gathering user data without the relevant person's knowledge or 'tricking' an individual into agreeing into sharing data. Whilst the first scenario can be perceived as being illegal, the second avoids this as they have obtained the user's consent. However, to what extent the user is conscious about having provided consent remains a question.

In many instances users do provide consent, but are unaware of the extent to which they have allowed capturing as well as the negative implications of 'carelessly' sharing data. Whilst the case of Cambridge Analytica using Facebook's data for their own agendas took the world by storm, it is interesting to note that many users willingly shared their, as well as their friends' data. Cambridge Analytica managed to harvest the information of an estimated 50 million users from a few hundred thousand persons who participated in one of many "free quizzes" on Facebook [1]. What is more disturbing are the findings of a 2017 research on 3,108 Massachusetts Institute of Technology (MIT) students, where 98% of participants shared their friends' email addresses in exchange for a free pizza [1]. This clearly shows that most individuals do not understand the value that data brings to organisations and more importantly, the negative implications of 'carelessly' sharing seemingly 'unimportant' data.

This research attempts to study the use of VPN clients amongst executive grade employees in Sri Lanka to determine their awareness levels on the potential security risks of VPN clients and its implications on data privacy.

II. LITERATURE REVIEW

Information security is broadly defined by [2] as "the protection of information and information systems from unauthorized access, use, disclosure, disruption, modification, or destruction in order to provide confidentiality, integrity, and availability". Ensuring security and privacy of information has now become a board-level concern for many organisations as well as individuals who possess information which they perceive to be important, valuable and 'worth protecting' [3].

Social Engineering, is essentially a technique to gain access to such 'valuable' information in an 'unauthorised' manner. Reference [4] defines Social Engineering as the "the art and science of getting people to comply with your wishes". In the context of this research, the 'wish' is to gain unauthorized access to information. Simply, Social Engineering tactics involve 'tricking' users into revealing valuable private information.

Technology plays a pivotal role in many social engineering schemes. Attack vectors include 'psychological manipulation' of potential victims, luring them into 'technology driven traps'. Six basic tendencies of human behavior have been identified, which could be 'exploited' by a would-be attacker to lure users into these traps. These are namely - authority, scarcity (e.g. limited time offers), reciprocation, consistency, liking, and social proof (e.g. extent of use of a technology among others). Reference [5] further adds 'inexperience' and 'curiosity' of users to the above list, which could allow them to become victims of social engineering.

978-1-5386-9136-6/18/$31.00 ©2018 IEEE

Authorized licensed use limited to: Qatar National Library. Downloaded on March 15,2026 at 11:00:04 UTC from IEEE Xplore. Restrictions apply.

In order to address such user-centric attack attempts, it is important to increase the level of awareness among users. Reference [6] defines Information security awareness as the general knowledge possessed by end-users on security practices and the resulting consequences of failure in following them.

The work of [3] highlights several Security-related behavioral theories such as:

- Theory of Reasoned Action – Individuals follow instructions/policies which define the non-misuse of information.
- General Deterrence Theory – Individuals do not misuse information as they are informed of the consequences.
- Others include the Protection Motivation Theory and Technology Acceptance Model.

Combined analysis of the above theories suggest that security awareness has a direct influence on user attitudes and behaviours [7] which would determine if they are likely to become victims of social engineering attempts. It should also be noted that the work of [8] which studies user awareness with regard to phishing shows that only a mere 36% were fully aware of how well phishing attacks are designed to increase their effectiveness. This lack of awareness can be considered as a key factor which could be adopted by individuals who wish to execute other Social Engineering attacks, beyond phishing.

Smartphones are mobile phones which have evolved ahead of their primary functionality – telecommunication. Not only do they function as multiple devices, they also possess processing capabilities on par with an average modern day computer, increasingly "mirroring functionality found on computers as a result of having modern mobile operating systems which support a wide range of general-purpose applications termed apps [9]". These mobile applications which can be downloaded within few minutes, vary from standard internet browsers, email clients and calendaring applications to more interactive and engaging virtual reality, social networking applications and mapping tools [10]. Due to these reasons, smartphones have become an integral part of our lives and function as the gateway to individuals' work and personal lives and are used to generate, process, store and transmit large volumes of work related and personal data.

The mobile device market is currently dominated by two mobile operating systems: Google's Android and Apple's iOS, both of which are supported by various VPN apps.

The technology behind Virtual Private Networks has been in existence for over 20 years. Reference [11] defines a VPN simply as "private network constructed within a public network infrastructure, such as the global Internet". The primary reasons for use of VPNs are privacy concerns among users who use public infrastructure to connect to the internet. These users need to ensure that no third-party can access the data while in transit without their explicit permission. Another common use of VPNs stems from the underlying technology. Using a VPN involves connecting to the VPN server which in-turn, communicates with the desired website. These VPN servers can be located anywhere around the world. Since requests to websites will originate from the VPN server, these can be easily used to circumvent geographic restrictions on certain websites. This includes the ability to access region specific content as well as bypass any government restrictions on websites.

However, as with any mobile app, these VPN apps should be used with caution. The Android OS in general bears an increased privacy risk [12], primarily due to the vast availability of apps, Android's open source coding platform as well as the large variety of available data [13]. Reference [14] states that apps which violate users' privacy can be easily developed and implemented within the Android ecosystem. It has also been identified that apps in the Google Playstore were found to leak sensitive information [15]. Further, a study by [16] revealed that malware in android devices actively harvested data of personal nature such as text messages, telephone contacts and user account information. Google has since introduced many security mechanisms to address these issues which include malware analysis on all apps in the Playstore as well as device level analysis of all apps which are installed. Nonetheless, [17] proves that these mechanisms are ineffective, with a success ratio of just 15%.

However, privacy violations can occur even in instances where a user knowingly allows a benign app (an app whose functions do not involve violating user privacy) to access confidential data. Such cases involve the app being used as a 'confused deputy' [18], where the app accidentally allows other malicious apps to use the permissions granted to it. Another instance is where apps are bundled with a malicious advertisement library [19]. Again, the permissions granted to the primary app are misused for malicious purposes. It should also be noted that many benign Android apps (including those which are completely legitimate) tend to request for more permissions than is actually needed to carry out their intended functionality [20].

Whilst both the iOS and Android mobile platforms adopt a permission based system which allows users to select which information, functionality and device features an application have access to, users have to respond to a myriad of permission requests, some of which are essential for certain apps. It is estimated that users make an average of 5 permission decisions per app [21] with the average mobile user installing 95 mobile apps on their devices [22]. The end result is that many users 'blindly' grant permission to various apps to 'speed up' the installation process. The work of [23] shows that users are mostly unaware of the degree to which they have allowed apps to access their confidential information and in many instances, are uncomfortable with the degree of access granted to various apps.

VPNs exist as both Paid and Free versions and are available as mobile applications as well to cater to the current smartphone users. As of 2016, there existed 283 VPN apps in the 1.4 million app Google Playstore [24]. However, findings suggest that these apps should be used with extreme caution as they display "suspicious behaviours". Reference [24] highlights some of these in detail as follows:

Android permissions and third-party user tracking: 75% of the VPN apps analysed during the study utilise tracking

978-1-5386-9136-6/18/$31.00 ©2018 IEEE

Authorized licensed use limited to: Qatar National Library. Downloaded on March 15,2026 at 11:00:04 UTC from IEEE Xplore. Restrictions apply.

libraries from third-parties. Further, 82% request access to sensitive information which includes user account information and text messages, permissions which would normally not be required to provide the promised services of the VPN.

Malware: The VPN apps in the study were uploaded to VirusTotal for analysis and over 38% showed to contain some malware presence. However, only a marginal number of users have highlighted any security or privacy related concerns in the app reviews. In fact, 25% of the apps in question were awarded a minimum of a 4-star rating.

It should be noted that this study was limited to free VPN apps due to the high costs involved in analyzing 'paid' apps. Reference [24] concludes that with the above deficiencies, millions of users may be subject to false guarantees of security and abusive practices inflicted by VPN apps, largely without their knowledge.

However, the studies carried out above have several shortcomings which are highlighted below:

- In most instances, the analysis focuses on the mobile apps and the permissions requested vis-a-vis their functionality. Focus on user awareness on permissions granted is minimal.
- Where users were actually involved and their awareness levels analyzed, it was observed that a significant proportion were teenagers/university students.
- Other conditions (e.g. sampling only gaming users, users whose devices were installed with a particular version of android) imposed restrictions on the sample used for the research.

It could be noted that teenagers/university students may install apps and grant them excessive permissions due to:

- Lack of awareness
- Possessing less critical information on their mobile devices
- Curiosity to try out unique functionalities of mobile apps, especially those complementing their lifestyles.

As such, this research aims to identify detailed information on the following aspects of mobile VPN client usage and awareness in Sri Lanka:

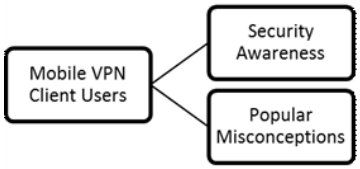

Fig. 1. Reasearch Aspects for Mobile VPN Client Users

The research aims to seek answers to the following questions:

- If VPN clients were used to access Facebook?
- What are the reasons for not using VPNs? (Awareness/Misconceptions)
- If VPN users are aware of the inherent risks? (Awareness)
- What are the reasons for using VPNs despite the awareness of the risks? (Misconceptions)

A key reason to conduct this research at this point in time was the sudden surge in use of VPNs in Sri Lanka following a temporary ban on social media platforms. In early March 2018, the Sri Lankan government instructed the Telecommunications Regulatory Commission of Sri Lanka to impose a ban on Facebook, WhatsApp, Twitter and other social media platforms after it emerged that posts containing 'hate speech' were being circulated, resulting in communal clashes in the central region of the country [25].

With social media platforms being immensely popular in Sri Lanka, users were quick to look for alternate means to access the blocked sites; and quickly found the 'key to all their problems' – VPNs. The Minister of Digital Infrastructure stated that over 800,000 VPN downloads had originated from Sri Lanka with an estimated 1 million users from Sri Lanka 'active' on despite the ban being in effect [25]. Whilst there were several articles highlighting the dangers of unknown VPN clients in Sri Lankan website, VPN providers were quick to capitalize on the situation and even offer special 'deals' for users from Sri Lanka. Guidelines on using VPNs, including YouTube videos surfaced quickly on the web, with some of them being in local languages.

Attempts to create some awareness on the downsides of using VPNs were noted [26][27]. However, these were few in number and do not appear to have reached out to the majority of users.

Thus, this research aims to identify Facebook usage via VPNs during the ban and the reasons for continuing with the use of VPNs, despite being made aware of the inherent risks.

III. Research Methodology

This study was carried out using a quantitative design. A quantitative design was adopted in carrying out the study with multiple choice answer questionnaires used to capture data.

A. Sample

Based on the formulae published by Krejcie & Morgan [28] to determine the sample size from a population, a sample of 152 persons was suggested. Studies by [29] show that the average rate of response to questionnaires of this nature among Sri Lankans is at 40%. Thus, the sample size was calculated at 380 persons, so that a 40% response rate would yield 152 responses. A pre-condition to be included in the sample was to be an active Facebook user (spending over 5 hours a week). A key shortcoming in the existing literature is that they focus on

978-1-5386-9136-6/18/$31.00 ©2018 IEEE

Authorized licensed use limited to: Qatar National Library. Downloaded on March 15,2026 at 11:00:04 UTC from IEEE Xplore. Restrictions apply.

teenagers, which is the reason for carrying out this research amongst executive grade employees.

B. Sampling Technique

Stratified Random Sampling was used for the purpose of selecting the sample with participants divided into strata on the basis of a single characteristic – gender (Male or Female).

IV. DATA COLLECTION

Questionnaires were used to carry out the data collection with Google Forms being used to capture responses and easily distribute among the sample. Links to the forms were shared via Facebook. Data collection was completed within a period of 2 weeks from 155 respondents.

The questionnaire was created based on the conceptual framework derived from the literature review. It aimed to identify the extent to which users are aware of security and privacy risks in VPN apps so that better awareness programs can be executed.

Part I of the questionnaire was designed to capture demographics such as gender and age group. Part II consisted of 4 multiple choice answer questions which were structured to capture the following information:

- Usage of mobile VPN clients
- Security Awareness of users/non-users
- Popular misconceptions among users/non-users

The questionnaire was reviewed by 2 persons in the Information Security domain and found to be suitable to gather the required data.

V. DATA ANALYSIS

The results from the questionnaire responses were analyzed and the information is indicated in the following tables and illustrations.

VI. RESULTS AND DISCUSSION

It was also observed that majority of the respondents (43.23%) fall within the 25-34 year age category with the 45-54 year and 55 year and above age categories contributing to only 9.68% of respondents combined. This trend can be easily attributed to the fact that the questionnaire was specifically shared amongst frequent and extensive Facebook users. It should also be noted that the 25-34 years and 35-44 year age categories represent a combined 62.58% of the sample. This clearly indicates that the majority of the respondents, whilst being relatively young, possibly possess sufficient experience and a level of maturity to understand the possible consequences of data leakages.

TABLE I. DEMOGRAPHICS

Age Group	Gender	
	Male	*Female*
18-24	19	24
25-34	30	37
35-44	16	14
45-54	9	6
55 and above	0	0
Total	74	81

Upon analysis of these findings, several conclusions can be derived with regards to user awareness on VPN clients. They are listed below as follows:

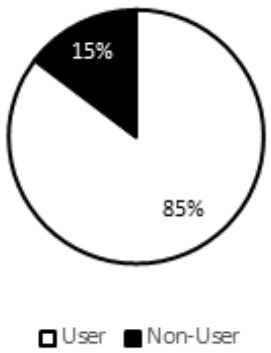

Fig. 2. Usage of VPN clients

TABLE II. REASONS FOR NON-USE OF VPN CLIENTS

Reason	Respondents	Percentage
Security	5	21.74 %
Legal	19	82.61 %
Other	7	30.43 %

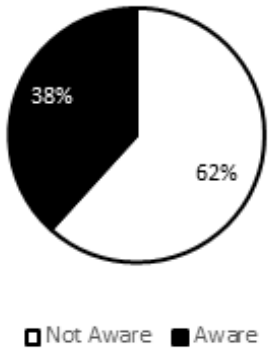

Fig. 3. Awareness on Possible Risks among VPN users

978-1-5386-9136-6/18/$31.00 ©2018 IEEE

Authorized licensed use limited to: Qatar National Library. Downloaded on March 15,2026 at 11:00:04 UTC from IEEE Xplore. Restrictions apply.

TABLE III. REASONS FOR USING VPNS DESPITE THE RISK

Reason	Respondents	Percentage
False information from government	43	86.96%
App permissions are sufficient	50	100.00%
Hosted data provider security is sufficient	33	66.00%
Data is anyway public, hence nothing to lose	29	58.00%

A. *Use of VPN clients to access Facebook*

Analysis of the responses to the first question revealed that 85% of respondents admitted to using some form of VPN client on their mobile devices to access Facebook, whilst the remaining 15% did not. This shows that a vast majority of the sample population did use VPNs to access Facebook.

B. *Reasons for Non-use of VPN clients*

Analysing the responses of the 15% (23 respondents) who did not use VPNs, 82.61% of the respondents cited fear of legal repercussions as a reason for not using a VPN client, 30.43% cited other reasons whilst only a mere 21.74% have cited security concerns as being a reason for not using a VPN. This implies that if user concerns on legality and other non-security aspects of using VPN clients were cleared, the number of VPN client users would have further increased.

C. *User Awareness on Risks of VPNs*

It was observed that from the 132 respondents who claimed to have used VPNs, 65.15% admitted that they were not aware of any risks in using VPNs. Nonetheless, this leaves a remaining 34.85% of users who have used VPNs despite being aware of possible security risks.

D. *Reasons for using VPNs despite the risk*

Analysing further the findings of 6.4, the last question attempts to identify the reasons as to why a staggering 34.85%, despite being aware that VPNs could compromise their confidential data, chose to use a VPN client to access Facebook. It should be noted, however, that the 50 respondents simply stated that they are AWARE of a POSSIBILITY of a security threat. This does not necessarily imply that they accept that the threat could be realized. Thus, the responses to this question identifies some of the popular misconceptions amongst VPN users with regard to possible security risks.

- 86.96% of users believe that 'fears' of security threats to data were being disseminated by the government and other law enforcement agencies to deter the use of VPNs, which defeats the purpose of the government restrictions.
- All respondents (100.00%) held the opinion that only granting the required permissions at the time of installing the VPN (e.g. a VPN client would not require Camera access) would ensure, at least to a great extent, that no privacy violations can occur.
- 66.00% of users were of the opinion that despite whatever possibilities of privacy violations by the VPN client, they would be countered by the 'advanced security controls' implemented by Facebook, which is where all data is stored.
- Interestingly, 58.00% of respondents stated that there is no guarantee of privacy for data hosted on Facebook and that it may already be available in the public domain. Hence use of a VPN client will not make any difference to the privacy of their data.

VII. CONCLUSION AND WAY FORWARD

Upon analysis of the findings, it can be noted that a majority (85%) of the sample did use VPNs during the period in which access to Facebook was limited. Analysis of respondents who did not use VPNs at all again show that only a minority (21.74%) refrained from using them due to security risks. This implies that if other non-security implications were not present, the number of VPN users would have increase.

The above findings substantiate the work of [24] which highlights the fact that an extremely low number of users have mentioned any form of security or privacy related issues in the app reviews, with 25% of the apps in question being awarded a 4-star rating. Low user-awareness on information security concerns is also highlighted by [8], with only 36% being fully aware of phishing tactics, further affirming the findings of this research.

It should be noted that due to time and resource constrains, this study was limited to users of Facebook. The ban which came into effect blocked access to all social media as well as instant messaging platforms (including, but not limited to Instagram, WhatsApp, Viber and Skype). Individuals who do not use Facebook use other social media and instant messaging platforms, both for personal and business purposes. Thus, the number of VPN users could be much greater than what is indicated, as this research focused specifically on active Facebook users.

Another aspect which could not be reviewed due to time and resource constrains is an analysis of the individual VPN apps used. Future research could focus on identifying the VPN apps used and carrying out an in-depth analysis to identify any privacy violating permissions.

Smartphone (and internet) users should be clearly informed of possible information security breaches that could occur as a result of using 'alternate technologies' such as VPNs. This should be carried out on a periodic basis and in proactive manner. Misconceptions amongst users identified in this research should be specifically addressed when carrying out such awareness campaigns. This would ensure that users have a lesser tendency to misinterpret security advice for government and vendor sponsored statements. Other elements such as education levels, how tech-savvy the sample is, the ways in which users spend their time on Facebook and information on how users became aware of VPN apps and its uses would also be helpful in preparing more effective awareness campaigns. Again, all such awareness should ideally be carried out in all three languages (Sinhala, Tamil & English). SMS/Email

978-1-5386-9136-6/18/$31.00 ©2018 IEEE

Authorized licensed use limited to: Qatar National Library. Downloaded on March 15,2026 at 11:00:04 UTC from IEEE Xplore. Restrictions apply.

campaigns, Facebook awareness campaigns, user targeted advertisements etc. are all suitable media for disseminating this awareness among the general public, for the betterment of individuals, organisations as well as the sovereignty of the nation.

REFERENCES

[1] K. Paul, "Americans are worried about their Facebook data — but college students would give up their privacy for free pizza." Internet: https://www.marketwatch.com/story/college-students-would-give-up-their-friends-privacy-for-free-pizza-2017-06-13 [18 Apr. 2018].

[2] R. Kissel, "Glossary of key information security terms," National Institute of Standards and Technology, Tech. Rep. NISTIR 7298 Revision 2, 2013.

[3] J. Ophoff and M. Robinson, "Exploring end-user smartphone security awareness within a South African context," 2014 Information Security for South Africa, Johannesburg, pp. 1-7, 2014

[4] Harl, "The Psychology of Social Engineering," Text of Harl's talk at Access All Areas III, May 1997

[5] M. Jordan, and H. Gouday, "The signs, and semiotics of the successful semantic attack." 14th Annual EICAR Conference (pp. 344-364). 2005.

[6] D.-L. Huang, P.-L. Patrick Rau, G. Salvendy, F. Gao, and J. Zhou, "Factors affecting perception of information security and their impacts on IT adoption and security practices," International Journal of Human-Computer Studies, vol. 69, no. 12, pp. 870–883, Dec. 2011.

[7] B. Bulgurcu, H. Cavusoglu, and I. Benbasat, "Information security policy compliance: An empirical study of rationality-based beliefs and information security awareness," MIS Q., vol. 34, no. 3, p. 523548, Sep. 2010.

[8] Jayatilleke, A. and Pathirana, P., "Why Phishing continues to work," International Journal of Scientific & Engineering Research, vol. 9, pp. 2265-2271, January 2018.

[9] T. Dorflinger, A. Voth, J. Kramer, and R. Fromm, "'My smartphone is a safe!' The user's point of view regarding novel authentication methods and gradual security levels on smartphones," in Proceedings of the 2010 International Conference on Security and Cryptography (SECRYPT), Jul. 2010, pp. 1–10.

[10] J. M. Urban, C. J. Hoofnagle, and S. Li, "Mobile phones and privacy," Social Science Research Network, Rochester, NY, SSRN Scholarly Paper ID 2103405, Jul. 2012.

[11] P. Ferguson, and G. Huston, "What is a VPN?," 1998.

[12] L. Marinos, A. Sfakianakis, "Enisa threat landscape." Technical report, ENISA (2012)

[13] M. Theoharidou, A. Mylonas, D. Gritzalis, "A risk assessment method for smartphones." SEC 2012. IFIP AICT, vol. 376, pp. 443–456. Springer, Heidelberg (2012)

[14] A. Mylonas, S. Dritsas, B. Tsoumas, D. Gritzalis, "Smartphone security evaluation – the malware attack case." Proceedings of International Conference of Security and Cryptography, pp. 25–36 (2011)

[15] W. Enck, P. Gilbert, B. Chun, L. Cox, J. Jung, P. McDaniel, A. Sheth, "TaintDroid: an information-flow tracking system for realtime privacy monitoring on smartphones." Proceedings of the 9th USENIX Conference on Operating Systems Design and Implementation, pp. 1–6. USENIX Association, Berkeley (2010)

[16] Y. Zhou, X. Jiang, "Dissecting android malware: characterization and evolution." Proceedings of the 2012 IEEE Symposium on Security and Privacy, pp. 95–109. IEEE Computer Society (2012)

[17] X. Jiang," An Evaluation of the Application Verification Service in Android 4.2." Csc2.ncsu.edu. Internet: https://www.csc2.ncsu.edu/faculty/xjiang4/appverify/ [11 Apr. 2018].

[18] E. Chin, A.P. Felt, K. Greenwood, D. Wagner, "Analyzing inter-application communication in Android." Proceedings of the 9th International Conference on Mobile Systems, Applications and Services, pp. 239–252. ACM, New York (2011)

[19] P. Pearce, A.P. Felt, G. Nunez, D. Wagner, "Android: privilege separation for applications and advertisers in android." Proceedings of the 7th ACM Symposium on Information, Computer and Communications Security, pp. 71–72. ACM (2012)

[20] A. Felt, E. Chin, S. Hanna, D. Song, D. Wagner, "Android permissions demystified." Proceedings of the 18th ACM Conference on Computer and Communications Security, pp. 627–638. ACM (2011)

[21] K. Olmstead, and M. Atkinson, "Chapter 3: An Analysis of Android App Permissions." Pew Research Center: Internet, Science & Tech. Internet: http://www.pewinternet.org/2015/11/10/an-analysis-of-android-app-permissions/ [12 Apr. 2018].

[22] P. Sawers, "Android Users Have an Average of 95 Apps Installed on Their Phones." The Next Web. Internet: https://thenextweb.com/apps/2014/08/26/android-users-average-95-apps-installed-phones-according-yahoo-aviate-data/ [16 Apr. 2018].

[23] R. Balebako, J. Jung, W. Lu, L. F. Cranor, and C. Nguyen, "Little brothers watching you: Raising awareness of data leaks on smartphones." Proc. SOUPS, 2013.

[24] M. Ikram, N. Vallina-Rodriguez, S. Seneviratne, MA Kaafar, V. Paxson, "An analysis of the privacy and security risks of android VPN permission-enabled apps." Proceedings of the 2016 Internet Measurement Conference (pp. 349-364), ACM. (2016)

[25] "Social Media ban to be lifted on Friday - Min. Harin Fernando - Sri Lanka News|News 1st|Sirasa News|Politics|Weather|Sports|Breaking News|Latest News". Internet: https://www.newsfirst.lk/2018/03/social-media-ban-lifted-friday-min-harin-fernando/ [18 April 2018].

[26] "How safe are free VPNs? – Sri Lanka News - Newsfirst | Breaking News and Latest News provider | Political | Sports | International | Business". Internet: https://www.newsfirst.lk/2018/03/safe-free-vpns/ [24 April 2018].

[27] "Those who used VPN to log into Fb at risk" - says SLCERT - Hiru News". Internet: http://www.hirunews.lk/186153/those-who-used-vpn-to-log-into-fb-at-risk-says-slcert [24 April 2018].

[28] R.V. Krejcie & D.W. Morgan, "Determining sample size for research activities." Educational and psychological measurement, 30(3), 607-610. (1970)

[29] Pathirana, P. and Azam, S.M.F., "Behavioural Intention to use Social Media among the Banking Consumers in Sri Lanka – An Empirical Study," European Journal of Social Sciences, vol. 54, pp. 123-133, February 2017.

978-1-5386-9136-6/18/$31.00 ©2018 IEEE

Authorized licensed use limited to: Qatar National Library. Downloaded on March 15,2026 at 11:00:04 UTC from IEEE Xplore. Restrictions apply.